

SIMATS ENGINEERING
ASSESSMENT TOOL 1
Experiments

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

SOFTWARE: CISCO PACKET TRACER, WIRESHARK

EXPERIMENTS

QUESTIONS:4

Total Marks:40

1. Design a simple LAN consisting of two PCs connected through a switch using Cisco Packet Tracer. Configure appropriate IP addresses for both hosts and verify connectivity using the ping command. Simultaneously capture the packet exchange using Wireshark. Identify the Ethernet frame fields such as Source MAC Address, Destination MAC Address, EtherType, and Frame Check Sequence (FCS). Explain how the Data Link layer uses MAC addressing to ensure successful frame delivery within the local network.
2. Create a network topology consisting of two LANs connected through a router in Cisco Packet Tracer. Configure IPv4 addressing, subnet masks, and default gateways for all devices, and verify end-to-end communication using the ping command. Capture the ICMP packets using Wireshark and analyze the IPv4 header fields, including Source IP Address, Destination IP Address, Time-To-Live (TTL), Protocol Number, and Header Length. Explain the role of the Network layer in forwarding packets between different networks.
3. Construct a local area network with multiple hosts connected through a switch in Cisco Packet Tracer. Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command. Capture the communication in Wireshark and identify the ARP Request and ARP Reply packets. Analyze the sender and target MAC/IP addresses and explain how the Address Resolution Protocol resolves logical IP addresses into physical MAC addresses before data transmission.
4. Develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer. Configure the required IP addresses and test connectivity using the ping command. Capture the ICMP traffic using Wireshark and identify the Echo Request and Echo Reply messages. Examine important ICMP fields such as Type, Code, Identifier, and Sequence Number, and calculate the Round Trip Time (RTT). Discuss how ICMP assists in network troubleshooting and connectivity verification.

5. Create a client-server network in Cisco Packet Tracer by configuring a web server and a client PC. Enable the HTTP service on the server and access the hosted webpage from the client using a web browser. Capture the communication in Wireshark and identify the TCP three-way handshake, followed by the HTTP GET request and HTTP response messages. Analyze the application-layer communication and explain how TCP provides reliable delivery for HTTP traffic through connection establishment, acknowledgments, and ordered data transfer.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding ; limited or partial integration	Poor understanding ; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

3. Construct a local area network with multiple hosts connected through a switch in

Cisco Packet Tracer. Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command. Capture the communication in Wireshark and identify the ARP Request and ARP Reply packets. Analyze the sender and target MAC/IP addresses and explain how the Address Resolution Protocol resolves logical IP addresses into physical MAC addresses before data transmission.

Aim

To construct a LAN using multiple hosts and a switch in Cisco Packet Tracer, clear the ARP cache, and analyze ARP Request and ARP Reply packets using Wireshark to understand how IP addresses are resolved into MAC addresses.

Software used

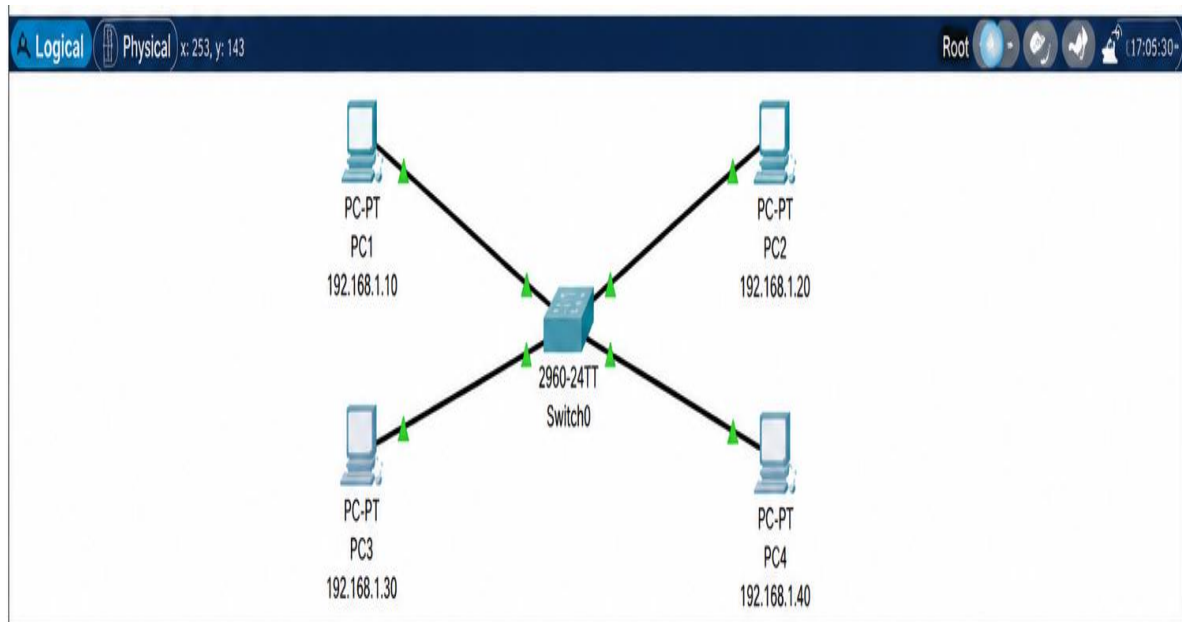
Cisco Packet Tracer, Wireshark

Procedure

1. Open Cisco Packet Tracer and create a new network topology.
2. Add multiple PCs and one Switch to the workspace.
3. Connect all PCs to the switch using Copper Straight-Through cables.
4. Configure suitable IPv4 addresses for the PCs, for example:
 - PC1: 192.168.1.10
 - PC2: 192.168.1.20
 - Subnet Mask: 255.255.255.0
5. Verify that all PCs are connected properly to the switch and that the interfaces are active.
6. Open the command prompt of PC1 and clear its ARP cache using the appropriate ARP command.
7. From PC1, execute the command ping 192.168.1.20 to initiate communication with PC2.
8. Open Wireshark and start capturing the network traffic on the appropriate network interface.
9. Apply the Wireshark filter arp and observe the ARP Request and ARP Reply packets. Identify the sender and target IP/MAC addresses.
10. Analyze the captured packets and explain that the ARP Request asks for the MAC address associated with the destination IP, while the ARP Reply provides that MAC address so the data can be transmitted through the local network.

Proof 1:

Construct a local area network with multiple hosts connected through a switch in Cisco Packet Tracer.



Proof 2:

Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command.

```
2. CISCO PACKET TRACER – SUCCESSFUL PING RESULT
PC1
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ping 192.168.1.20
Pinging 192.168.1.20 with 32 bytes of data:
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>|
```

Proof 3:

Capture the communication in Wireshark and identify the ARP Request and ARP Reply packets.

3. WIRESHARK – ARP REQUEST AND ARP REPLY PACKETS						
*Ethernet						
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help						
arp						
No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	Cisco_96:34:6d	Broadcast	ARP	42	Who has 192.168.1.20? Tell 192.168.1.10
2	0.000737	Cisco_1a:2b:3c	Cisco_96:34:6d	ARP	42	192.168.1.20 is at 00e0.f7e6.1a2b

> Frame 1: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface 0 > Ethernet II, Src: Cisco_96:34:6d (00e0.f7a6.963d.6d), Dst: Broadcast (ff:ff:ff:ff:ff:ff) > Address Resolution Protocol (request) - Hardware type: Ethernet (1) - Protocol type: IPv4 (0x0800) - Hardware size: 6 - Protocol size: 4 - Opcode: request (1) - Sender MAC address: 00e0.f7a6.963d.6d - Sender IP address: 192.168.1.10 - Target MAC address: 00:00:00:00:00:00 - Target IP address: 192.168.1.20	> Frame 2: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface 0 > Ethernet II, Src: Cisco_1a:2b:3c (00e0.f7e6.1a2b), Dst: Cisco_96:34:6d (00e0.f7a6.963d.6d) > Address Resolution Protocol (reply) - Hardware type: Ethernet (1) - Protocol type: IPv4 (0x0800) - Hardware size: 6 - Protocol size: 4 - Opcode: reply (2) - Sender MAC address: 00e0.f7e6.1a2b - Sender IP address: 192.168.1.20 - Target MAC address: 00e0.f7a6.963d.6d - Target IP address: 192.168.1.10
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Final Explanation

The LAN was successfully created using Cisco Packet Tracer, with multiple PCs connected through a switch and configured with IP addresses in the same network. Connectivity was verified successfully using the ping command.

After clearing the ARP cache, communication between the PCs generated an ARP Request followed by an ARP Reply. The ARP Request was used to find the MAC address associated with the destination IP address, while the ARP Reply provided the required MAC address.

The packets were captured and analyzed using Wireshark, where the sender and target IP/MAC addresses and ARP request/reply information were observed. Thus, the experiment demonstrates how ARP resolves an IP address into a MAC address before data transmission within a local network.

4. Develop a simple network consisting of two PCs connected through a router in

Cisco Packet Tracer. Configure the required IP addresses and test connectivity

using the ping command. Capture the ICMP traffic using Wireshark and identify

the Echo Request and Echo Reply messages. Examine important ICMP fields

such as Type, Code, Identifier, and Sequence Number, and calculate the Round

Trip Time (RTT). Discuss how ICMP assists in network troubleshooting and connectivity verification.

Aim

To create a network with two PCs connected through a router using Cisco Packet Tracer, verify connectivity using ping, and analyze ICMP Echo Request and Echo Reply packets using Wireshark, including RTT calculation.

Software used

Cisco packet tracer, Wireshark.

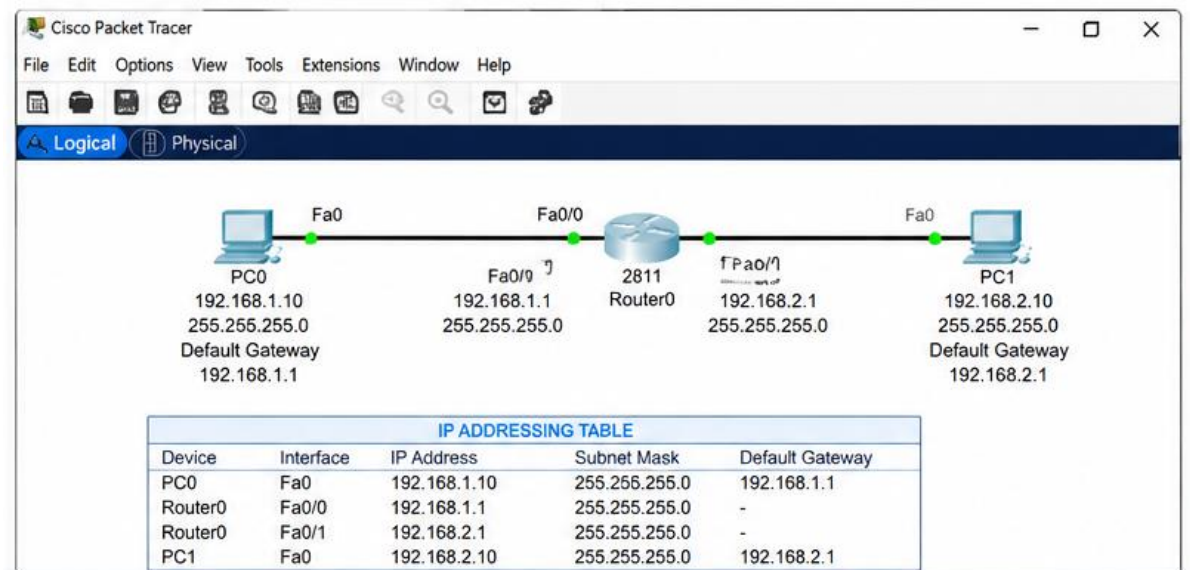
Procedure

1. Open Cisco Packet Tracer and create a new network topology.
2. Add two PCs and one router to the workspace.
3. Connect PC1 to the router and PC2 to the router using suitable Ethernet cables.
4. Configure the IP address and subnet mask for PC1 and the router interface connected to PC1.
5. Configure the IP address and subnet mask for PC2 and the router interface connected to PC2. Set the appropriate default gateway on both PCs.
6. Turn on/configure the required router interfaces and verify that the network links are active.
7. Open the command prompt on PC1 and execute the `ping` command using the IP address of PC2.
8. Open Wireshark, start packet capture, and perform the ping again to capture the ICMP traffic.
9. Apply the Wireshark filter `icmp` and identify the Echo Request and Echo Reply packets. Examine the Type, Code, Identifier, and Sequence Number fields.
10. Record the request and reply times and calculate the Round Trip Time (RTT) as the time taken for an ICMP Echo Request to travel to the destination and the corresponding Echo Reply to return. Use the result to verify connectivity and understand how ICMP helps in network troubleshooting.

Proof 1:

Develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer.

1. CISCO PACKET TRACER – NETWORK TOPOLOGY



Proof 2:

Configure the required IP addresses and test connectivity using the ping command. Capture the ICMP traffic using Wireshark and identify the Echo Request and Echo Reply messages.

2. SUCCESSFUL PING RESULT (CONNECTIVITY TEST)

```
Command Prompt

C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Proof 3: Examine important ICMP fields such as Type, Code, Identifier, and Sequence Number, and calculate the Round Trip Time (RTT). Discuss how ICMP assists in network troubleshooting and connectivity verification.

3. WIRESHARK – ICMP ECHO REQUEST AND ECHO REPLY ANALYSIS

Capturing from Ethernet

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter: <Ctrl>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.1.10	192.168.2.10	ICMP	98	Echo (ping) request id=0x1234, seq=1/256, ttl=128
2	0.000786	192.168.2.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0x1234, seq=1/256, ttl=128
3	1.001102	192.168.1.10	192.168.2.10	ICMP	98	Echo (ping) request id=0x1234, seq=2/512, ttl=128
4	1.001891	192.168.2.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0x1234, seq=2/512, ttl=128
5	2.002163	192.168.1.10	192.168.2.10	ICMP	98	Echo (ping) request id=0x1234, seq=3/768, ttl=128
6	2.003030	192.168.2.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0x1234, seq=3/768, ttl=128

> Frame 1: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface \Device\NPF_{...}

> Ethernet II, Src: 0c:4b:54:00:01:00 (0c:4b:54:00:01:00), Dst: 00:50:79:66:68:00 (00:50:79:66:68:00)

> Internet Protocol Version 4, Src: 192.168.1.10, Dst: 192.168.2.10

Internet Control Message Protocol

- Type: 8 (Echo (ping) request)
- Code: 0
- Checksum: 0x7cfd [correct]
- [Checksum Status: Good]
- Identifier (BE): 0x1234 (4660)
- Identifier (LE): 0x3412 (13330)
- Sequence Number (BE): 1 (0x0001)
- Sequence Number (LE): 256 (0x0100)

> Frame 2: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface \Device\NPF_{...}

> Ethernet II, Src: 00:50:79:66:68:00 (00:50:79:66:68:00), Dst: 0c:4b:54:00:01:00 (0c:4b:54:00:01:00)

> Internet Protocol Version 4, Src: 192.168.2.10, Dst: 192.168.1.10

Internet Control Message Protocol

- Type: 0 (Echo (ping) reply)
- Code: 0
- Checksum: 0x7cfd [correct]
- [Checksum Status: Good]
- Identifier (BE): 0x1234 (4660)
- Identifier (LE): 0x3412 (13330)
- Sequence Number (BE): 1 (0x0001)
- Sequence Number (LE): 256 (0x0100)

IMPORTANT ICMP FIELDS

(From Echo Request Packet)

- **Type:** 8 (Echo Request)
- **Code:** 0
- **Identifier:** 0x1234 (4660)
- **Sequence Number:** 1 (0x0001)

ROUND TRIP TIME (RTT) CALCULATION

Echo Request (Packet 1) Time: 0.000000 sec

Echo Reply (Packet 2) Time: 0.000786 sec

RTT = 0.000786 – 0.000000

RTT = 0.000786 sec = 0.786 ms

Final explanation:

The two PCs were successfully connected through the router and communication was verified using ping. Wireshark captured the ICMP Echo Request sent by the source PC and the Echo Reply returned by the destination PC. The ICMP **Type**, **Code**, **Identifier**, and **Sequence Number** fields were examined, and the RTT was calculated from the request/reply timestamps. ICMP is useful for checking host reachability, measuring network delay, and identifying connectivity problems.